

BANG NA AGROMET, Thailand

WMO: 484530

Lat: 13.6664N

Lon: 100.6061E

Elev: 6

StdP: 101.25

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.9	21.5	12.7	9.2	22.6	14.5	10.3	24.1	4.6	27.1	4.3	27.6	0.9	50	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	7.0	36.4	27.6	35.6	27.4	34.8	27.1	28.7	33.7	28.3	33.3	28.0	32.8	3.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	23.6	31.3	27.1	22.9	30.9	26.7	22.3	30.6	93.4	33.8	91.5	33.3	89.5	32.9	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.5	4.2	3.8	17.7	37.4	1.8	1.1	16.5	38.2	15.4	38.8	14.4	39.4	13.1	40.3		
(5)			14.4	29.0	2.2	0.6	12.8	29.5	11.6	29.8	10.4	30.2	8.8	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	27.2	28.7	30.0	31.2	31.0	30.2	29.6	29.6	29.3	29.0	28.8	27.5	(6)
(7)		DBStd	1.87	1.95	1.46	1.66	1.61	1.58	1.11	1.10	0.95	1.08	1.13	1.32	2.06	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	7052	533	524	619	635	650	606	607	607	578	588	563	541	(10)
(11)		CDD18.3	4010	275	291	361	385	392	356	348	348	328	330	313	283	(11)
(12)		CDH23.3	49477	2878	3386	4628	5244	5356	4709	4384	4344	3904	3861	3732	3050	(12)
(13)		CDH26.7	22728	1072	1415	2259	2902	2911	2347	1979	1943	1642	1547	1541	1171	(13)
(14)	Wind	WSAvg	1.3	1.0	1.4	1.8	1.7	1.5	1.4	1.3	1.3	1.1	0.8	1.1	1.2	(14)
(15)	Precipitation	PrecAvg	1442	18	19	46	79	187	169	154	184	293	244	39	9	(15)
(16)		PrecMax	1865	77	78	155	235	373	349	255	355	487	420	145	53	(16)
(17)		PrecMin	977	0	0	3	9	54	54	36	81	169	49	0	0	(17)
(18)		PrecStd	223	21	21	42	61	80	73	61	80	73	83	41	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	34.6	35.7	37.2	37.3	36.2	35.1	35.2	34.8	34.3	34.2	34.4	(19)
(20)			MCWB	24.6	25.2	26.1	27.4	28.0	27.5	27.3	26.8	26.7	26.5	25.8	25.5	(20)
(21)		2%	DB	33.0	33.5	34.8	36.5	36.4	34.8	34.0	34.0	34.0	33.6	33.5	33.3	(21)
(22)			MCWB	24.8	25.4	26.5	27.6	27.9	27.4	26.9	26.8	26.8	26.5	25.7	25.2	(22)
(23)		5%	DB	32.2	32.8	34.0	35.6	35.6	34.0	33.3	33.1	33.1	32.9	32.9	32.5	(23)
(24)			MCWB	24.2	25.5	26.8	27.6	27.8	27.2	26.7	26.6	26.7	26.4	25.5	24.9	(24)
(25)		10%	DB	31.3	32.1	33.3	34.8	34.5	33.1	32.5	32.4	32.4	32.1	32.2	31.5	(25)
(26)			MCWB	23.9	25.3	26.5	27.4	27.5	26.9	26.5	26.4	26.6	26.2	25.4	24.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.4	28.1	29.4	29.4	28.7	27.8	27.7	28.1	27.8	27.7	27.1	(27)
(28)			MADB	31.5	32.2	33.5	34.7	34.8	33.4	32.8	32.6	32.8	32.1	31.8	31.6	(28)
(29)		2%	WB	26.4	26.9	27.6	28.8	28.7	28.1	27.5	27.4	27.5	27.3	27.1	26.4	(29)
(30)			MADB	30.3	31.4	32.8	33.8	33.7	32.8	32.2	31.9	32.0	31.3	31.3	30.6	(30)
(31)		5%	WB	26.0	26.5	27.3	28.4	28.3	27.7	27.1	27.1	27.2	27.0	26.7	25.9	(31)
(32)			MADB	29.5	30.8	32.2	33.3	33.2	32.2	31.6	31.4	31.5	30.9	30.7	30.0	(32)
(33)		10%	WB	25.5	26.2	26.9	28.0	27.9	27.3	26.8	26.7	26.8	26.7	26.3	25.5	(33)
(34)			MADB	28.9	30.1	31.7	32.8	32.7	31.7	31.0	30.8	30.9	30.3	30.0	29.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	7.2	6.6	7.0	6.7	6.1	5.9	6.1	6.4	6.3	6.5	7.4	(35)
(36)			MCDBR	8.8	7.9	7.4	7.7	7.7	7.3	6.9	7.0	7.3	7.2	7.3	7.9	(36)
(37)			MCWBR	2.6	2.3	1.9	2.3	2.4	2.4	2.3	2.4	2.4	2.2	2.2	2.4	(37)
(38)		5% WB	MADB	7.0	6.7	6.7	7.0	7.2	6.8	6.3	6.4	6.8	6.6	6.7	7.2	(38)
(39)			MCWBR	2.3	2.0	2.0	2.3	2.5	2.5	2.3	2.3	2.6	2.3	2.2	2.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.544	0.610	0.635	0.574	0.498	0.466	0.459	0.460	0.484	0.577	0.524	0.509	(40)	
(41)		taud	1.963	1.831	1.805	1.992	2.261	2.349	2.369	2.351	2.275	1.972	2.099	2.088	(41)	
(42)		Ebn at Noon	756	725	720	763	809	827	835	843	829	742	770	776	(42)	
(43)		Edh at Noon	180	213	223	185	139	125	123	127	138	183	156	156	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.10	5.54	5.68	6.07	5.53	5.27	5.04	5.10	4.93	4.78	4.93	4.91	(44)	
(45)		RadStd	0.29	0.23	0.42	0.38	0.39	0.28	0.26	0.32	0.28	0.26	0.34	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	+0.20	+0.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page